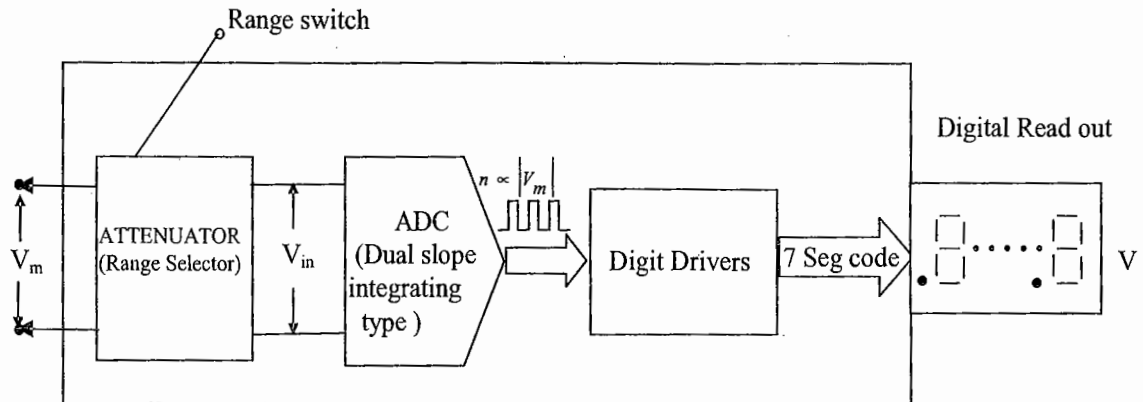


Digital Voltmeters

- A Digital Voltmeter is a Voltage measuring unit whose block diagram is shown below:



Analog DC voltage to be measured is first converted into pulses by Analog-to-Digital converter. Such Digital output is then driven to a digital display via Digit Drivers.

- The most preferred ADC in the design of Digital meters is Dual Slope Integrating type ADC since it offers High Accuracy, High Noise rejection, High Stability and Less Hardware complexity than other types of ADCs (Non Integrating types)
- Advantages of Digital meters over Analog meters:**
 - High input Impedance
 - Less Loading Error
 - No Observational Errors, No frictional Errors
 - Better Accuracy
 - Superior resolution
 - Fast operating speed
 - Easy readability
 - Programmability
 - Computerized control
 - Auto Ranging and Over Ranging facilities

Block diagram of digital multi meter is shown below:

